# DIY Spectrometer

*MATERIALS*

- 1 CD, DVD, or diffraction grating
- 1 paper roll
- Cardboard/paper (to fill the ends of the paper roll)
- Masking Tape & Packing Tape
- Scissors & X-Acto Knife/box cutter
- Pencil

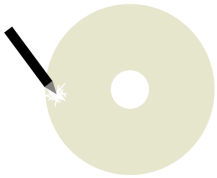
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*INSTRUCTIONS*

**Step 1: Preparing the CD/DVD/Diffraction Grating**

*Option A: CD Preparation (easier)*

- Use an X-Acto Knife to scrape off a tiny section of the thin cover near the edge of the CD.



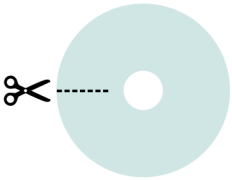
- Take a piece of packing tape and stick it onto the section that you just scraped off. Peel the tape off and the cover should come off with the tape.



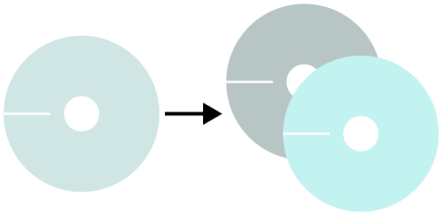
- Repeat this for the entire CD until the cover is fully removed.

*Option B: DVD Preparation (harder)*

- Take a pair of scissors and make a cut towards the center of the DVD. The cut does not have to go all the way through (see below)



- Then from the cut that you just made, carefully split the DVD into its two layers with your fingernails. The layer that we want is the clear layer (without the top sticker)!



Option C: Diffraction grating preparation (easy)

- Using a pencil, trace a circle around the diffraction grating with the roll and cut with scissors.

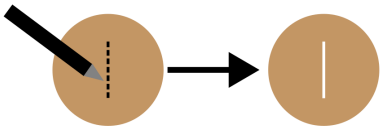


**Step 2: Preparing the ends of the spectroscope**

a) Use your pencil to trace out 2 circles with the paper roll. Cut both circles out with your scissors.



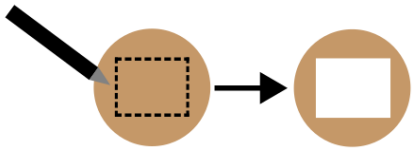
b) Circle #1: Cut a thin slit in the middle of the circle with your X-Acto knife or box cutter.



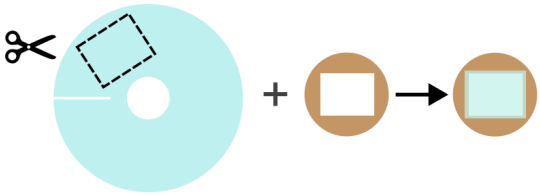
c) Circle #2 will be the diffraction grating end.

i) If you're using a CD/DVD:

1) Draw a rectangle inside the circle (the corners should not reach the edges!) with a pencil. Cut out the rectangle with an X-Acto knife/box cutter for the window.



2) Cut a section of the CD/DVD big enough to cover the window and tape it to the window. Use masking tape to cover up the sharp edges.



ii) If you're using a pre-bought diffraction grating, you can simple use the cutout from Step 1 Option C (or use cardboard/paper to cover up the edges if you want)

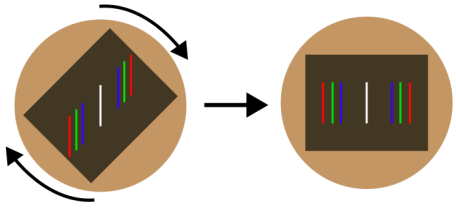
**Step 3: Assembling the spectroscope**

a) Tape the circle with the slit to one end of the paper roll with masking tape.



b) Hold up the spectroscope and diffraction grating and view from the diffraction grating side. Point it towards a light source **(not the sun!)** and you should see some rainbow lines!

c) Rotate the diffraction grating so that the lines are clearly split across from the slit (see below).



d) Once you find the right orientation, tape it to the paper roll.

